

DBMS – PAYMENT MANAGEMENT

Project Title: Etsy Handmade Marketplace – Payment Management System

Skill Task: DBMS

Week: 5

1. Database Design

1.1 Payment Table

Attribute	Data Type	Constraint
<i>Payment_ID</i>	NUMBER(5)	Primary Key
<i>Order_ID</i>	NUMBER(5)	Foreign Key
<i>Payment_Mode</i>	VARCHAR2(20)	NOT NULL
<i>Payment_Date</i>	DATE	NOT NULL
<i>Payment_Amount</i>	NUMBER(10,2)	NOT NULL
<i>Payment_Status</i>	VARCHAR2(20)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID      NUMBER(5) PRIMARY KEY,  
3     Order_ID        NUMBER(5),  
4     Payment_Mode     VARCHAR2(20) NOT NULL,  
5     Payment_Date     DATE NOT NULL,  
6     Payment_Amount   NUMBER(10,2) NOT NULL,  
7     Payment_Status   VARCHAR2(20) NOT NULL,  
8     CONSTRAINT fk_payment_order  
9         FOREIGN KEY (Order_ID)  
10        REFERENCES Orders(Order_ID)  
11 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (501, 2001, 'UPI', SYSDATE, 1450.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (502, 2002, 'Card', SYSDATE, 890.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (503, 2003, 'Net Banking', SYSDATE, 1800.00, 'Failed');
```

1 row created.

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)  
3 VALUES  
4 (504, 2004, 'UPI', SYSDATE, 650.00, 'Successful');
```

1 row created.

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
501	2001	UPI	28-SEP-26	1450
502	2002	Card	28-SEP-26	890
504	2004	UPI	28-SEP-26	650

5.2 Display Failed Payments

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	2003	Net Banking	28-SEP-26	1800

6. Modify Payment Status

```
SQL> UPDATE Payment
  2  SET Payment_Status = 'Successful'
  3  WHERE Payment_ID = 503;
```

1 row updated.

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_ID = 503;
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	2003	Net Banking	28-SEP-26	1800

7. Analyze Payment Methods

Count the transactions made through each payment mode

```
SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_TRANSACTIONS
-----	-----
Net Banking	1
Card	1
UPI	2

Calculate amount collected by payment mode

```
SQL> SELECT
  2     Payment_Mode,
  3     SUM(Payment_Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_AMOUNT
-----	-----
Net Banking	1800
Card	890
UPI	2100

8. Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     o.Order_ID,
  4     c.Customer_ID,
  5     c.Customer_Name,
  6     p.Payment_Mode,
  7     p.Payment_Date,
  8     p.Payment_Amount,
  9     p.Payment_Status
 10 FROM Payment p
 11 JOIN Orders o
 12     ON p.Order_ID = o.Order_ID
 13 JOIN Customer c
 14     ON o.Customer_ID = c.Customer_ID
 15 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
504	2004	204	Ava Martinez	UPI	28-SEP-26	650	Successful
503	2003	203	Liam Turner	Net Banking	28-SEP-26	1800	Successful
502	2002	202	Sophia Clark	Card	28-SEP-26	890	Successful
501	2001	201	Emma Wilson	UPI	28-SEP-26	1450	Successful

4 rows selected.

```
SQL> COMMIT;

Commit complete.
```